

Experiment-5

Interactive Dashboard Development using Tableau / Microsoft Power BI

```
In [1]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Step 1: Load Dataset
df = pd.read_csv(
    "Sample - Superstore.csv",
    encoding="latin1"
)
print("Dataset Loaded Successfully")
print(df.head())

# Step 2: Check Dataset
print("\nDataset Shape:")
print(df.shape)
print("\nColumn Names:")
print(df.columns.tolist())

# Step 3: Verify Data
print("\nData Types:")
print(df.dtypes)
print("\nMissing Values:")
print(df.isnull().sum())

# Step 4: Calculate KPIs
total_sales = df["Sales"].sum()
total_profit = df["Profit"].sum()
total_orders = df["Order ID"].nunique()
average_order_value = total_sales / total_orders
print("\n===== KPIs =====")
print("Total Sales:", round(total_sales, 2))
print("Total Profit:", round(total_profit, 2))
print("Total Orders:", total_orders)
print("Average Order Value:", round(average_order_value, 2))

# Step 5: Bar Chart - Sales by Region
region_sales = df.groupby("Region")["Sales"].sum()
plt.figure(figsize=(8, 5))
region_sales.plot(kind="bar")
plt.title("Sales by Region")
plt.xlabel("Region")
plt.ylabel("Total Sales")
plt.xticks(rotation=0)
plt.tight_layout()
plt.show()

# Step 6: Line Chart - Monthly Sales
df["Order Date"] = pd.to_datetime(
    df["Order Date"],
    format="mixed",
    errors="coerce"
)
monthly_sales = df.groupby(
    df["Order Date"].dt.to_period("M")
```

```

)["Sales"].sum()
plt.figure(figsize=(10, 5))
monthly_sales.plot(
    kind="line",
    marker="o"
)
plt.title("Monthly Sales Trend")
plt.xlabel("Month")
plt.ylabel("Sales")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

# Step 7: Pie Chart - Sales by Category
category_sales = df.groupby("Category")["Sales"].sum()
plt.figure(figsize=(7, 7))
category_sales.plot(
    kind="pie",
    autopct="%1.1f%%"
)
plt.title("Sales by Category")
plt.ylabel("")
plt.tight_layout()
plt.show()

# Step 8: Correlation Heatmap
numeric_data = df.select_dtypes(
    include="number"
)
correlation = numeric_data.corr()
plt.figure(figsize=(9, 6))
sns.heatmap(
    correlation,
    annot=True,
    fmt=".2f",
    cmap="coolwarm"
)
plt.title("Correlation Heatmap")
plt.tight_layout()
plt.show()

# Step 9: Filter / Slicer
print("\nAvailable Regions:")
print(df["Region"].unique())
region = input(
    "\nEnter Region (East/West/North/South): "
)
filtered_data = df[
    df["Region"].str.lower() == region.lower()
]
print("\n===== Filtered Data =====")
print(filtered_data.head())
filtered_sales = filtered_data["Sales"].sum()
filtered_profit = filtered_data["Profit"].sum()
print("\nFiltered Sales:",
      round(filtered_sales, 2))
print("Filtered Profit:",
      round(filtered_profit, 2))

# Step 10: Business Insights

```

```
print("\n===== Business Insights =====")
top_region = region_sales.idxmax()
top_category = category_sales.idxmax()
print("Top Sales Region:", top_region)
print(
    "Highest Region Sales:",
    round(region_sales.max(), 2)
)
print("Top Sales Category:", top_category)
print(
    "Highest Category Sales:",
    round(category_sales.max(), 2)
)
```

Dataset Loaded Successfully

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	\
0	1	CA-2016-152156	11-08-2016	11-11-2016	Second Class	CG-12520	
1	2	CA-2016-152156	11-08-2016	11-11-2016	Second Class	CG-12520	
2	3	CA-2016-138688	06-12-2016	6/16/2016	Second Class	DV-13045	
3	4	US-2015-108966	10-11-2015	10/18/2015	Standard Class	SO-20335	
4	5	US-2015-108966	10-11-2015	10/18/2015	Standard Class	SO-20335	

	Customer Name	Segment	Country	City	...	\
0	Claire Gute	Consumer	United States	Henderson	...	
1	Claire Gute	Consumer	United States	Henderson	...	
2	Darrin Van Huff	Corporate	United States	Los Angeles	...	
3	Sean O'Donnell	Consumer	United States	Fort Lauderdale	...	
4	Sean O'Donnell	Consumer	United States	Fort Lauderdale	...	

	Postal Code	Region	Product ID	Category	Sub-Category	\
0	42420	South	FUR-BO-10001798	Furniture	Bookcases	
1	42420	South	FUR-CH-10000454	Furniture	Chairs	
2	90036	West	OFF-LA-10000240	Office Supplies	Labels	
3	33311	South	FUR-TA-10000577	Furniture	Tables	
4	33311	South	OFF-ST-10000760	Office Supplies	Storage	

	Product Name	Sales	Quantity	\
0	Bush Somerset Collection Bookcase	261.9600	2	
1	Hon Deluxe Fabric Upholstered Stacking Chairs,...	731.9400	3	
2	Self-Adhesive Address Labels for Typewriters b...	14.6200	2	
3	Bretford CR4500 Series Slim Rectangular Table	957.5775	5	
4	Eldon Fold 'N Roll Cart System	22.3680	2	

	Discount	Profit
0	0.00	41.9136
1	0.00	219.5820
2	0.00	6.8714
3	0.45	-383.0310
4	0.20	2.5164

[5 rows x 21 columns]

Dataset Shape:
(9994, 21)

Column Names:

['Row ID', 'Order ID', 'Order Date', 'Ship Date', 'Ship Mode', 'Customer ID', 'Customer Name', 'Segment', 'Country', 'City', 'State', 'Postal Code', 'Region', 'Product ID', 'Category', 'Sub-Category', 'Product Name', 'Sales', 'Quantity', 'Discount', 'Profit']

Data Types:

Row ID	int64
Order ID	object
Order Date	object
Ship Date	object
Ship Mode	object
Customer ID	object
Customer Name	object
Segment	object
Country	object
City	object
State	object
Postal Code	int64

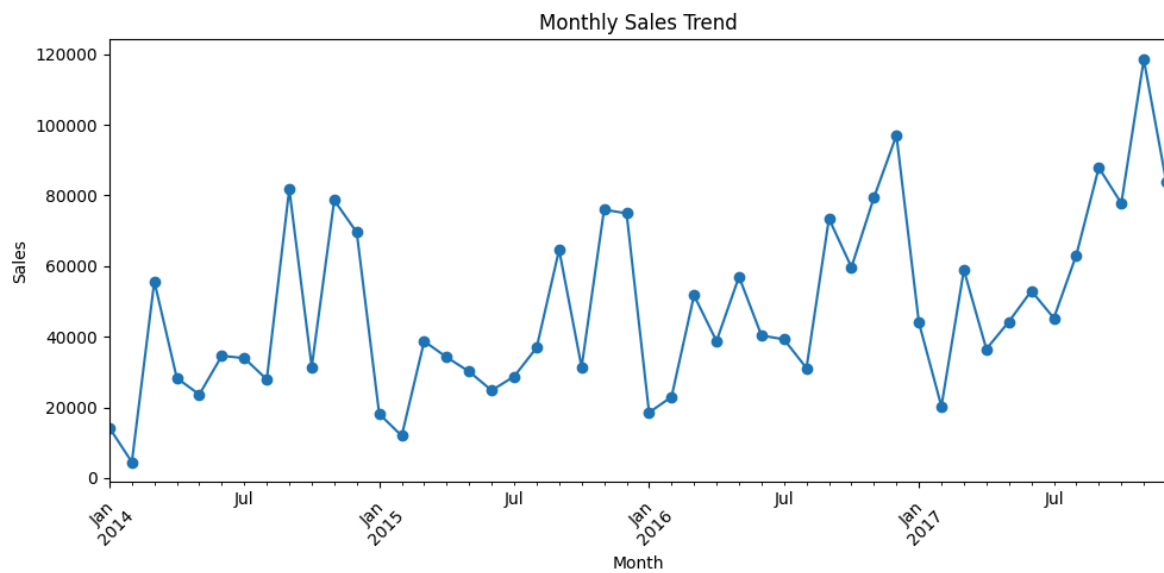
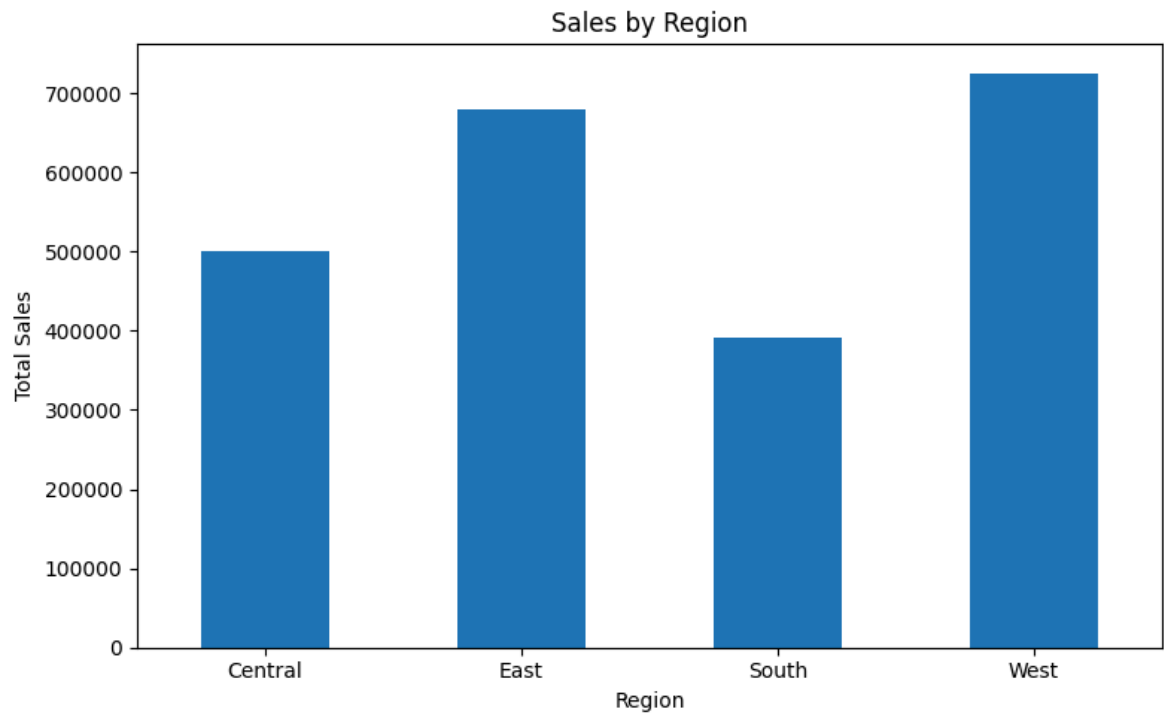
```
Region          object
Product ID      object
Category        object
Sub-Category    object
Product Name    object
Sales           float64
Quantity        int64
Discount        float64
Profit          float64
dtype: object
```

Missing Values:

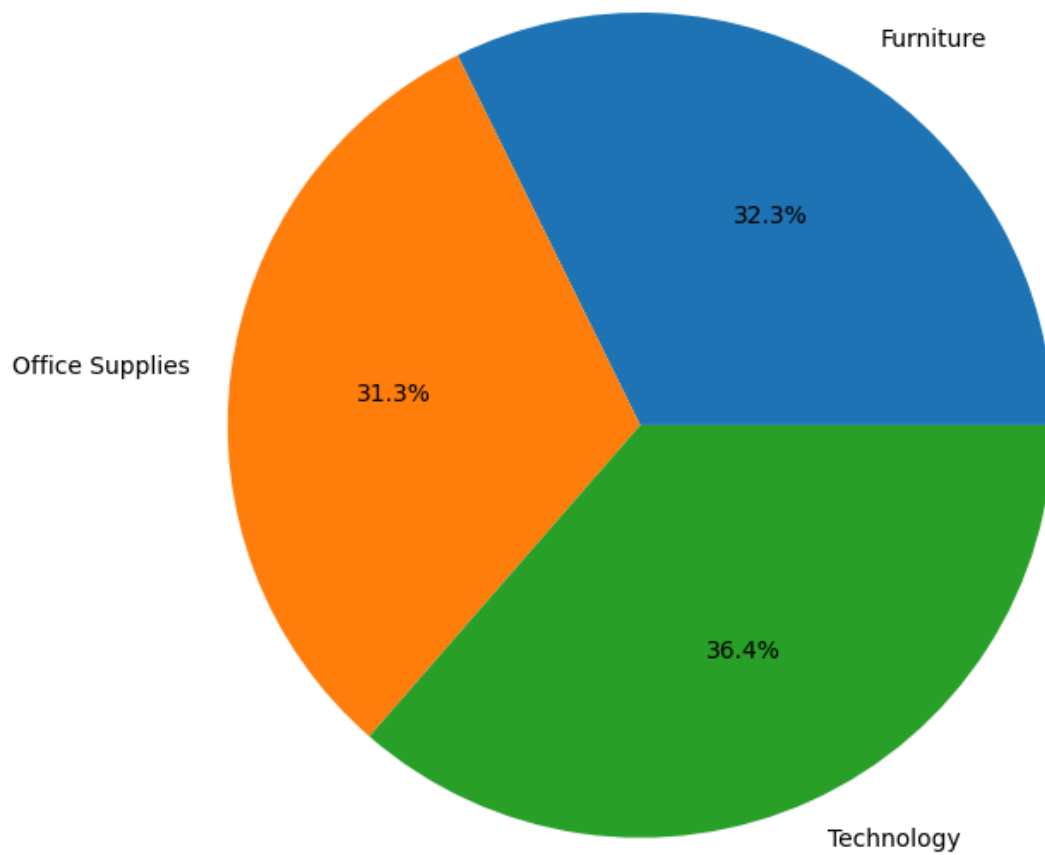
```
Row ID          0
Order ID        0
Order Date      0
Ship Date       0
Ship Mode       0
Customer ID     0
Customer Name   0
Segment         0
Country         0
City            0
State           0
Postal Code     0
Region          0
Product ID      0
Category        0
Sub-Category    0
Product Name    0
Sales           0
Quantity        0
Discount        0
Profit          0
dtype: int64
```

===== KPIs =====

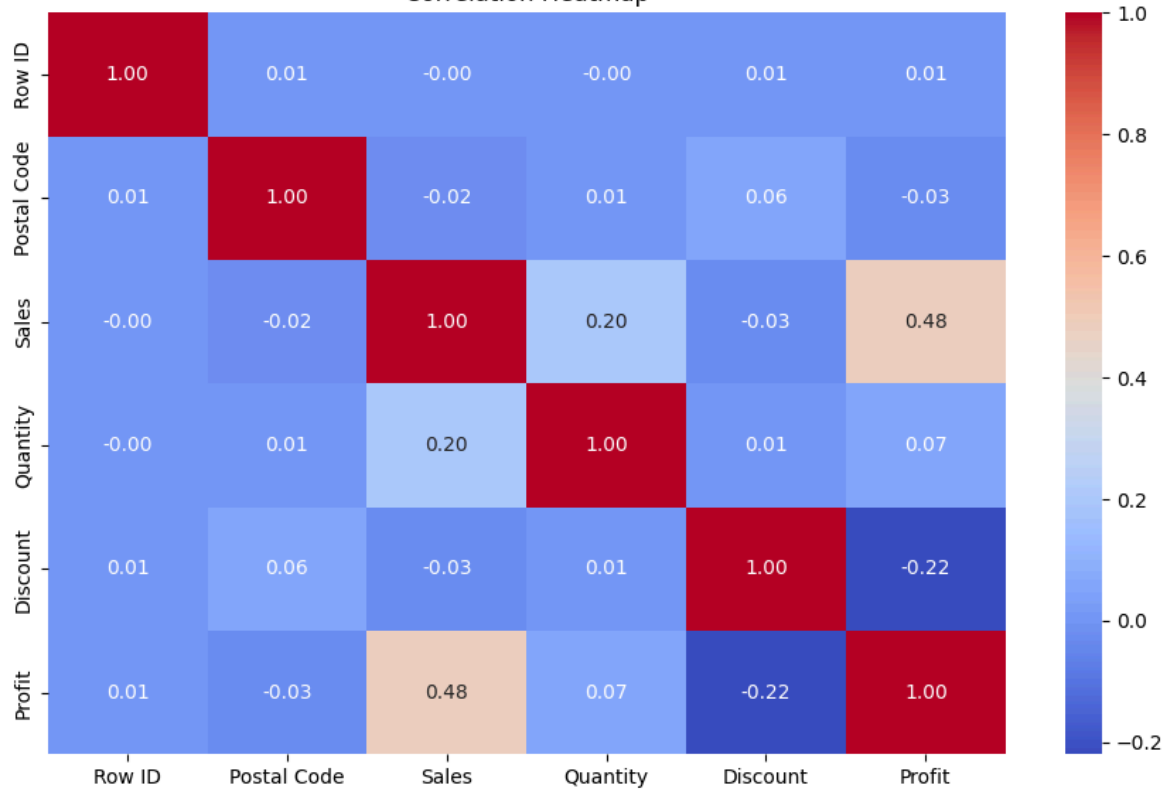
```
Total Sales: 2297200.86
Total Profit: 286397.02
Total Orders: 5009
Average Order Value: 458.61
```



Sales by Category



Correlation Heatmap



Available Regions:

['South' 'West' 'Central' 'East']

===== Filtered Data =====

Empty DataFrame

Columns: [Row ID, Order ID, Order Date, Ship Date, Ship Mode, Customer ID, Customer Name, Segment, Country, City, State, Postal Code, Region, Product ID, Category, Sub-Category, Product Name, Sales, Quantity, Discount, Profit]

Index: []

[0 rows x 21 columns]

Filtered Sales: 0.0

Filtered Profit: 0.0

===== Business Insights =====

Top Sales Region: West

Highest Region Sales: 725457.82

Top Sales Category: Technology

Highest Category Sales: 836154.03

Questions and Answers

1. What is Business Intelligence (BI)? How does it support organizational decision-making?

Business Intelligence (BI) is a technology-driven process used to collect, analyze, and visualize business data. It helps organizations make better decisions by providing meaningful insights into business performance, trends, and customer behavior.

2. What is an interactive dashboard? How is it different from a static report?

An **interactive dashboard** is a visual interface that allows users to explore data using filters, slicers, charts, and drill-down options.

- **Interactive Dashboard:** Users can interact with and explore data dynamically.
- **Static Report:** Displays fixed information and usually does not allow users to interact with the data.

3. Compare Tableau and Microsoft Power BI based on features, usability, and business applications.

Feature	Tableau	Microsoft Power BI
Visualization	Strong and highly customizable	Strong and easy to use
Usability	User-friendly for data visualization	User-friendly, especially for Microsoft users
Data Sources	Supports many data sources	Supports many data sources
Integration	Integrates with various platforms	Strong integration with Microsoft products
Business Applications	Advanced analytics and visualization	Business reporting, dashboards, and analytics

Both tools are widely used for **data analysis, reporting, visualization, and business decision-making**.

4. What are Key Performance Indicators (KPIs)? Give examples of KPIs used in sales analytics.

Key Performance Indicators (KPIs) are measurable values used to evaluate the performance of an organization, department, or business activity.

Examples of sales KPIs include:

- **Total Sales**
- **Sales Growth Rate**
- **Profit Margin**
- **Number of Orders**
- **Average Order Value**
- **Customer Acquisition Cost**
- **Conversion Rate**

5. Explain the purpose of filters, slicers, and drill-down functionality in dashboards.

- **Filters:** Display only the data that matches selected conditions.
- **Slicers:** Provide an interactive visual way to filter dashboard data.
- **Drill-down:** Allows users to move from summarized information to more detailed information.

For example, a sales dashboard can be filtered by **year, region, category, or product**.

6. What factors should be considered while designing an effective business dashboard?

Important factors include:

- Clear and simple layout.
- Relevant KPIs and metrics.
- Appropriate chart types.
- Accurate and reliable data.
- Easy-to-understand labels.
- Consistent formatting.
- Interactive filters and drill-downs.
- Fast loading and good performance.
- Avoiding unnecessary visual elements.

7. How can dashboards help managers monitor organizational performance in real time?

Dashboards can connect to frequently updated data sources and display current business metrics. Managers can monitor **sales, revenue, inventory, customer activity, and other KPIs** in one place and quickly identify changes or unusual trends.

8. Why is data visualization important in Business Intelligence tools?

Data visualization converts complex data into **charts, graphs, maps, and other visual forms**. It makes patterns, trends, comparisons, and outliers easier to understand and helps users communicate data-driven insights effectively.

9. What insights can be derived from a sales dashboard for an e-commerce company?

A sales dashboard can provide insights such as:

- Total sales and revenue.
- Best-selling products.
- Sales by region or location.
- Monthly and yearly sales trends.
- Customer purchasing behavior.
- Average order value.
- Product profitability.
- Order and cancellation trends.
- Sales performance by category.
- Conversion and customer retention trends.

10. Explain how interactive dashboards improve communication between data analysts and business stakeholders.

Interactive dashboards provide a common visual platform where analysts and stakeholders can explore the same data. Stakeholders can apply filters, examine trends, and drill into specific details without requesting separate reports. This improves **understanding, collaboration, transparency, and data-driven decision-making**.